

Muhammad Abubakar

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SUMMARY

Final-year Cyber Security undergraduate at GIKI specializing in Application Security and DevSecOps, with internship experience in security testing and secure development at Thingtrax. Hands-on across threat modeling, SAST/SCA pipeline gating, and OWASP-based hardening, plus applied ML and wireless-security research.

EDUCATION

Ghulam Ishaq Khan Institute (GIKI) — BSc Cyber Security

Sep 2023 – Jun 2027

CGPA: 3.3 / 4.00 · 2× Dean's Honor List (5th & 6th semesters)

Relevant coursework: Information Security, Information Assurance (NIST / ISO 27001 / GRC), Secure Software Design, Mobile & Wireless Security, Computer Networks, Data Structures & Algorithms.

TECHNICAL SKILLS

Security: Application Security, DevSecOps, Penetration Testing, Vulnerability Assessment, Threat Modeling, OWASP Top 10, SAST/SCA

Frameworks & Standards: NIST CSF, ISO 27001, Secure SDLC

Tools: Burp Suite, Nmap, Metasploit, Wireshark, Bandit, pip-audit, GitHub Actions, Linux (Kali / Ubuntu)

Programming: Python, C++, JavaScript, SQL, Bash

Web, Data & ML: Flask, React, REST APIs, PostgreSQL · scikit-learn, TensorFlow, CatBoost, Pandas

PROFESSIONAL EXPERIENCE

Thingtrax, Lahore — Software QA Intern (Security Testing)

Jun 2025 – Aug 2025

- Designed and ran security-focused test cases across 4 production releases — auth-flow validation, input fuzzing, and dependency-vulnerability checks alongside functional and regression coverage.
- Triaged defects with developers and helped enforce the team's secure-coding bar; contributed to a digitization workflow that cut document retrieval from ~15 min to under 5 sec.

Thingtrax, Lahore — Secure Software Developer Intern

Jun 2024 – Aug 2024

- Built core flows (auth, catalog, checkout) for an industry-sponsored e-commerce platform using input validation and parameterized queries.
- Coordinated weekly agile syncs for a 10-person team and shipped 50+ UI screens informed by research on 200+ users, improving delivery cadence ~15%.

PROJECTS

Secure DevSecOps CI/CD Pipeline — AppSec Automation

[GitHub](#)

- Enforced fail-fast security gates in GitHub Actions (pytest, Bandit SAST, pip-audit SCA), blocking any push containing secrets, SQLi patterns, or vulnerable dependencies.
- Packaged it as a reusable template so downstream repos inherit the same security posture with zero per-project setup.

Bakri Pay — Secure Banking Application — Flask / AppSec

[GitHub](#)

- Full-stack Flask (MVC) banking app with secure session auth, transaction handling, and rules-based fraud detection on suspicious transfers.
- Mapped controls to OWASP Top 10 (bcrypt/Argon2 hashing, CSRF tokens, parameterized queries); validated via Burp Suite assessment.

Phantom Identity — Browser Fingerprint Privacy Extension — JavaScript

[GitHub](#)

- Manifest V3 extension that defeats passive fingerprint tracking by spoofing Canvas, WebGL, Navigator, Screen, and timezone attributes per session.
- Surfaces fingerprint entropy and spoof effectiveness in an in-extension dashboard; runs locally with zero outbound telemetry.

DQN-Enhanced Adaptive Artificial Noise Allocation — Wireless Security / RL

[GitHub](#)

- Modeled an eavesdropper-present MISO wiretap channel (MRT beamforming, artificial noise, Rayleigh fading, imperfect CSI) and trained a DQN agent to adaptively split power between signal and jamming noise — +0.299 bits/s/Hz secrecy gain over fixed allocation at low SNR.

AI-Integrated Intrusion Detection System — ML / Cybersecurity

[GitHub](#)

- Hybrid CatBoost + LOF detector pairing supervised classification with unsupervised novelty detection to catch unseen attacks; trained on 1.6M samples in ~7.5 min.

CERTIFICATIONS

- Google Cybersecurity Professional Certificate (Coursera, 2025) ·
- AI for Cyber Security (Macquarie, 2025)
- Linux Essentials (LPI) ·
- Cisco Networking Basics & Intro to Cybersecurity (Cisco, 2025)

